

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $A = \{\theta : \sin(\theta) = \tan(\theta)\}$ and $B = \{\theta : \cos(\theta) = 1\}$ be two sets. Then -

(A) $A = B$

(B) $A \subset B$ and $B - A \neq \phi$

(C) $A \not\subset B$

(D) $B \not\subset A$

MTE003

2. If $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = 3$, then the general solution of θ is -

(A) $2n\pi \pm \frac{\pi}{6}$

(B) $n\pi \pm \frac{\pi}{6}$

(C) $2n\pi \pm \frac{\pi}{3}$

(D) $n\pi \pm \frac{\pi}{3}$

where $n \in I$

MTE002

3. The solution set of $(5 + 4 \cos \theta)(2 \cos \theta + 1) = 0$ in the interval $[0, 2\pi]$ is :

(A) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$

(B) $\left\{\frac{\pi}{3}, \pi\right\}$

(C) $\left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$

(D) $\left\{\frac{2\pi}{3}, \frac{5\pi}{3}\right\}$

MTE004

4. The number of solutions of the equation $\sin 2x - 2 \cos x + 4 \sin x = 4$ in the interval $[0, 5\pi]$ is -

(A) 6

(B) 4

(C) 3

(D) 5

MTE007

5. If $2 \tan^2 \theta = \sec^2 \theta$, then the general solution of θ -

(A) $n\pi + \frac{\pi}{4} (n \in I)$

(B) $n\pi - \frac{\pi}{4} (n \in I)$

(C) $n\pi \pm \frac{\pi}{4} (n \in I)$

(D) $2n\pi \pm \frac{\pi}{4} (n \in I)$

MTE001

6. The general solution of equation $4 \cos^2 x + 6 \sin^2 x = 5$ is -

(A) $x = n\pi \pm \frac{\pi}{2} (n \in I)$

(B) $x = n\pi \pm \frac{\pi}{4} (n \in I)$

(C) $x = n\pi \pm \frac{3\pi}{2} (n \in I)$

(D) None of these

MTE005

7. If $\tan^2 \theta - (1 + \sqrt{3}) \tan \theta + \sqrt{3} = 0$, then the general value of θ is :

(A) $n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{3}$

(B) $n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{3}$

(C) $n\pi + \frac{\pi}{4}, n\pi - \frac{\pi}{3}$

(D) $n\pi - \frac{\pi}{4}, n\pi - \frac{\pi}{3}$

where $n \in I$

MTE006

Trigonometric Equations

8. The set of angles between 0 and 2π satisfying the equation $4 \cos^2 \theta - 2\sqrt{2} \cos \theta - 1 = 0$ is

(A) $\left\{ \frac{\pi}{12}, \frac{5\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12} \right\}$ (B) $\left\{ \frac{\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12} \right\}$
 (C) $\left\{ \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12} \right\}$ (D) $\left\{ \frac{\pi}{12}, \frac{7\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12} \right\}$

MTE008

9. If $\frac{\tan 2\theta + \tan \theta}{1 - \tan \theta \tan 2\theta} = 0$, then the general value of θ is -

(A) $n\pi; n \in I$ (B) $\frac{n\pi}{3}; n \in I$
 (C) $\frac{n\pi}{4}$ (D) $\frac{n\pi}{6}; n \in I$

where $n \in I$

MTE010

10. If $\tan \theta + \tan 4\theta + \tan 7\theta = \tan \theta \tan 4\theta \tan 7\theta$, then $\theta =$

(A) $\frac{n\pi}{4}$ (B) $\frac{n\pi}{7}$
 (C) $\frac{n\pi}{12}, n \neq (12m - 6), n, m \in I$ (D) $n\pi$

where $n \in I$

MTE009

11. If $\tan \theta - \sqrt{2} \sec \theta = \sqrt{3}$, then the general solution of θ is -

(A) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$ (B) $n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{4}$
 (C) $n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{4}$ (D) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}$

where $n \in I$

MTE011

12. The most general values of x for which $\sin x + \cos x = \min_{a \in R} \{1, a^2 - 4a + 6\}$ is given by -

(A) $2n\pi$ (B) $2n\pi + \frac{\pi}{2}$
 (C) $n\pi + (-1)^n \cdot \frac{\pi}{4} - \frac{\pi}{4}$ (D) None of these

where $n \in I$

MTE012

13. Number of values of x satisfying the equation $\log_2(\sin x) + \log_{1/2}(-\cos x) = 0$ in the interval $(-\pi, \pi]$ is equal to -

(A) 0 (B) 1 (C) 2 (D) 3

MTE014

14. If the equation $\sin^4 x - (k + 2) \sin^2 x - (k + 3) = 0$ has a solution then k must lie in the interval:

(A) $(-4, -2)$ (B) $[-3, 2)$ (C) $(-4, -3)$ (D) $[-3, -2]$

MTE013

15. If $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$, then the greatest positive solution of $1 + \sin^4 x = \cos^2 3x$ is -
 (A) π (B) 2π (C) $\frac{5\pi}{2}$ (D) none of these

MTE015

16. The number of solutions of the equation $\sin x = x^2 + x + 1$ is -
 (A) 0 (B) 1 (C) 2 (D) None

MTE016

17. The number of solutions of the equation $2\cos\left(\frac{x}{2}\right) = 3^x + 3^{-x}$ is -
 (A) 1 (B) 2 (C) 3 (D) None

MTE017

18. The equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + x^{-2}, 0 < x \leq \frac{\pi}{2}$ has
 (A) one real solutions (B) more than one real solutions
 (C) no real solution (D) none of the above

MTE018

19. Given $a^2 + 2a + \operatorname{cosec}^2\left(\frac{\pi}{2}(a+x)\right) = 0$ then, which of the following holds good ?
 (A) $a = 1; \frac{x}{2} \in I$ (B) $a = -1; \frac{x}{2} \in I$
 (C) $a \in R; x \in \phi$ (D) a, x are finite but not possible to find

MTE019

20. The complete solution set of the inequality $\tan^2 x - 2\sqrt{2}\tan x + 1 \leq 0$ is -
 (A) $n\pi + \frac{\pi}{8} \leq x \leq \frac{3\pi}{8} + n\pi, n \in I$ (B) $n\pi + \frac{\pi}{4} \leq x \leq \frac{3\pi}{4} + n\pi, n \in I$
 (C) $n\pi + \frac{\pi}{16} \leq x \leq \frac{3\pi}{8} + n\pi, n \in I$ (D) $n\pi + \frac{\pi}{3} \leq x \leq \frac{2\pi}{3} + n\pi, n \in I$

MTE020

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The equation $2\sin\frac{x}{2} \cdot \cos^2 x + \sin^2 x = 2\sin\frac{x}{2} \cdot \sin^2 x + \cos^2 x$ has a root for which
 (A) $\sin 2x = 1$ (B) $\sin 2x = -1$
 (C) $\cos x = \frac{1}{2}$ (D) $\cos 2x = -\frac{1}{2}$ MTE041
2. Which of the following set of values of x satisfies the equation $2^{(2\sin^2 x - 3\sin x + 1)} + 2^{(2 - 2\sin^2 x + 3\sin x)} = 9$ is -
 (A) $x = x\pi, n \in I$ (B) $x = 2n\pi + \frac{\pi}{2}, n \in I$
 (C) $x = m\pi \pm \frac{\pi}{6}, n \in I$ (D) $x = n\pi \pm \frac{\pi}{3}, n \in I$ MTE042
3. If sum of all the solution of the equation $\cot x + \operatorname{cosec} x + \sec x = \tan x$ in $[0, 2\pi]$ is $\frac{k\pi}{2}$, then the value of k is greater than
 (A) 1 (B) 2 (C) 3 (D) 4 MTE043
4. $\cos 4x \cos 8x - \cos 5x \cos 9x = 0$ if
 (A) $\cos 12x = \cos 14x$ (B) $\sin 13x = 0$
 (C) $\sin x = 0$ (D) $\cos x = 0$ MTE048
5. The equation $\sin x + \cos(k + x) + \cos(k - x) = 2$ has real solution(s), then $\sin k$ can be
 (A) $\frac{-3}{4}$ (B) $\frac{1}{4}$
 (C) $\frac{1}{2}$ (D) $\frac{3}{4}$ MTE044
6. The equation $\sin^6 x + \cos^6 x = a^2$ has real solution if
 (A) $a \in (-1, 1)$ (B) $a \in \left(-1, -\frac{1}{2}\right)$
 (C) $a \in \left(-1, \frac{1}{2}\right)$ (D) $a \in \left(\frac{1}{2}, 1\right)$ MTE047
7. If $m > 0, n < 5, 0 < m + n < 10$ and $\frac{x^2 + 5}{2} = x - 2 \cos(m + nx)$ has atleast one real root, then
 (A) the greatest value of $(m + n)$ is 3π .
 (B) the greatest value of $(m + n)$ is 2π .
 (C) the least value of $(m + n)$ is π .
 (D) the least value of $(m + n)$ is 2π . MTE045

8. If $x^4 + 3 \cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with $a, b, c \in (2, 5)$ then
 (A) $a + b + c = \pi$ (B) $a - b + c = \pi$
 (C) $a + b + c = 3\pi$ (D) $a - b + c = 3\pi$ MTE046
9. If the equation $\cot^4 x - 2 \operatorname{cosec}^2 x + a^2 = 0$ has atleast one solution then, sum of all possible integral values of 'a' is not equal to
 (A) 1 (B) 0
 (C) 3 (D) 2 MTE126

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 10 & 11

Let $f(x) = \cos x + \sin x - 1$ and $g(x) = \sin 2x - 2$.

10. Number of solutions of the equation $f(x) = g(x)$ in $x \in \left[-\pi, \frac{7\pi}{2}\right]$ is equal to
 (A) 4 (B) 5
 (C) 6 (D) 8 MTE049
11. If the equation $f(x) = k$ has atleast one real solution then the number of possible integral values of k is equal to
 (A) 1 (B) 2
 (C) 3 (D) 4 MTE050

Paragraph for Question No. 12 & 13

Let $f(x) = \sin^2 x - (a - 1) \sin x + 2(a - 3)$

On the basis of above information, answer the following questions :

12. If $x \in [0, \pi]$ and $f(x) = 0$ has exactly one real root, then 'a' lies in
 (A) (3, 5) (B) (2, 4)
 (C) (4, 5) (D) none of these MTE051
13. If $f(x) = 0$ have two real roots in $(0, \pi)$, then $a \in$
 (A) (1, 2) (B) (3, 4)
 (C) $(3, 4) \cup \{5\}$ (D) (3, 5) MTE052

MATCHING LIST TYPE QUESTION

- 14.**
- | List-I | | List-II |
|--|---|----------------|
| (I) Number of common solutions of the equations
$2 \cos^2 x - 3 \cos x + 1 = 0$ and $\tan\left(\frac{3x}{4}\right) + 1 = 0$,
where $-\pi < x \leq 3\pi$, is equal to | (P) | 2 |
| (II) Number of solutions of the equation
$\sin^2 x + 2 \sin x - 4 \cos x - \sin 2x = 0$ in $(0, 2\pi)$ is equal to | (Q) | 12 |
| (III) If the sum of all possible values of $x \in (0, 2\pi)$ satisfying
the equation $2 \cos x \operatorname{cosec} x - 4 \cos x - \operatorname{cosec} x = -2$
is equal to $\frac{k\pi}{4}$ ($k \in N$), then the value of k is | (R) | 4 |
| (IV) Let $x - y = \frac{1}{4}$ and $\cos(\pi x) \cdot \cos(\pi y) = \frac{1}{2\sqrt{2}}$
where $x, y \in (0, 2)$ then the number of ordered pair(s) (x, y)
is equal to | (S) | 3 |
| (A) $I \rightarrow P$; $II \rightarrow P$; $III \rightarrow Q$; $IV \rightarrow R$ | (B) $I \rightarrow T$; $II \rightarrow P$; $III \rightarrow S$; $IV \rightarrow R$ | |
| (C) $I \rightarrow R$; $II \rightarrow P$; $III \rightarrow S$; $IV \rightarrow T$ | (D) $I \rightarrow Q$; $II \rightarrow P$; $III \rightarrow S$; $IV \rightarrow T$ | |

MTE053

MATRIX MATCH TYPE QUESTION

- 15.**
- | Column I | | Column II |
|---|-----|------------------|
| (A) The number of real roots of the equation
$\cos^7 x + \sin^4 x = 1$ in $(-\pi, \pi)$ is | (P) | 1 |
| (B) The number of solutions of the equation
$ \cot x = \cot x + \frac{1}{\sin x}$ ($0 < x < \pi$) is | (Q) | 2 |
| | (R) | 0 |
| (C) The number of solution(s) of equation in $(0, 3\pi)$
$\sin^2 x + 3 \sec^2 x = -(\tan^2 x + \cot^2 x)$ is | (S) | 3 |
| (D) The number of values of $x \in [-2\pi, 2\pi]$, which
satisfy $\operatorname{cosec} x = 1 + \cot x$ | (T) | 5 |

MTE054

MTE055

MTE056

MTE057

EXERCISE (S)

1. Number of principal solution(s) of the equation $4 \cdot 16^{\sin^2 x} = 2^{6 \sin x}$ is MTE021
2. The number of solutions of the equation $\tan^2 x - \sec^{10} x + 1 = 0$ in $(0, 10)$ is - MTE022
3. If α, β ($\alpha - \beta \neq 2n\pi, n \in I$) are different values of θ satisfying the equation $5 \cos \theta - 12 \sin \theta = 11$.
If the value of $\sin(\alpha + \beta) = -\frac{5k}{169}$, then the value of k is MTE059
4. Find the number of solutions of the equation $\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1$ in the interval $[0, 2\pi]$. MTE071
5. In $(0, 6\pi)$, find the number of solutions of the equation $\tan \theta + \tan 2\theta + \tan 3\theta = \tan \theta \cdot \tan 2\theta \cdot \tan 3\theta$ MTE070
6. If $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $4 \cos^2 \left(\frac{\pi}{4} - \frac{x}{2}\right) + \sqrt{4 \sin^4 x + \sin^2 2x}$ is always equal to MTE060
7. If $\sin \pi x + \cos \pi x = 0$, then the sum of all solutions in $[0, 100]$ is MTE058
8. If the arithmetic mean of the roots of the equation $4 \cos^3 x - 4 \cos^2 x - \cos(\pi + x) - 1 = 0$ in the interval $[0, 315]$ is equal to $k\pi$, then find the value of k MTE061
9. If $0 \leq x \leq 3\pi, 0 \leq y \leq 3\pi$ and $\cos x \cdot \sin y = 1$, then the possible number of values of the ordered pair (x, y) is - MTE023
10. Number of integral solution(s) of the inequality $2 \sin^2 x - 5 \sin x + 2 > 0$ in $x \in [0, 2\pi]$, is - MTE025

EXERCISE - JEE (Main) PYQ

1. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are: [AIEEE 2011]
 (1) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$ (2) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$
 (3) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$ (4) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$
- MTE026
2. In a ΔPQR , if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$, then the angle R is equal to : [AIEEE 2012]
 (1) $\frac{3\pi}{4}$ (2) $\frac{5\pi}{6}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{4}$
- MTE027
3. If $0 \leq x < 2\pi$, then the number of real values of x , which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is :- [JEE (Main) 2016]
 (1) 9 (2) 3 (3) 5 (4) 7
- MTE028
4. If sum of all the solutions of the equation $8\cos x \cdot \left(\cos\left(\frac{\pi}{6} + x\right) \cdot \cos\left(\frac{\pi}{6} - x\right) - \frac{1}{2} \right) = 1$ in $[0, \pi]$ is $k\pi$, then k is equal to : [JEE (Main) 2018]
 (1) $\frac{13}{9}$ (2) $\frac{8}{9}$ (3) $\frac{20}{9}$ (4) $\frac{2}{3}$
- MTE029
5. The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is : [JEE (Main) 2019]
 (1) 5 (2) 4 (3) 7 (4) 3
- MTE030
6. Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to : [JEE (Main) 2019]
 (1) $[2, 6]$ (2) $[3, 7]$ (3) R (4) $[1, 4]$
- MTE031
7. If $\sin^4 \alpha + 4\cos^4 \beta + 2 = 4\sqrt{2} \sin \alpha \cos \beta$; $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to : [JEE (Main) 2019]
 (1) 0 (2) $-\sqrt{2}$ (3) -1 (4) $\sqrt{2}$
- MTE032
8. If the equation $\cos^4 \theta + \sin^4 \theta + \lambda = 0$ has real solutions for θ , then λ lies in the interval : [JEE (Main) 2020]
 (1) $\left[-\frac{3}{2}, -\frac{5}{4}\right]$ (2) $\left[-\frac{1}{2}, -\frac{1}{4}\right]$ (3) $\left[-\frac{5}{4}, -1\right]$ (4) $\left[-1, -\frac{1}{2}\right]$
- MTE033

9. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in : [JEE (Main) 2021]
 (1) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$ (2) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$
 (3) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$ (4) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$
 MTE034
10. If n is the number of solutions of the equation $2\cos x \left(4\sin\left(\frac{\pi}{4} + x\right)\sin\left(\frac{\pi}{4} - x\right) - 1\right) = 1, x \in [0, \pi]$ and S is the sum of all these solutions, then the ordered pair (n, S) is : [JEE (Main) 2021]
 (1) $(3, 13\pi/9)$ (2) $(2, 2\pi/3)$ (3) $(2, 8\pi/9)$ (4) $(3, 5\pi/3)$
 MTE035
11. The number of roots of the equation, $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ in the interval $[0, \pi]$ is equal to : [JEE (Main) 2021]
 (1) 3 (2) 4 (3) 8 (4) 2
 MTE036
12. The number of solutions of the equation $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x, x \in [-3\pi, 3\pi]$ is : [JEE (Main) 2022]
 (1) 8 (2) 5 (3) 6 (4) 7
 MTE037
13. Let $S = \left\{\theta \in [-\pi, \pi] - \left\{\pm \frac{\pi}{2}\right\} : \sin \theta \tan \theta + \tan \theta = \sin 2\theta\right\}$. If $T = \sum_{\theta \in S} \cos 2\theta$, then $T + n(S)$ is equal [JEE (Main) 2022]
 (1) $7 + \sqrt{3}$ (2) 9 (3) $8 + \sqrt{3}$ (4) 10
 MTE038
14. Let $S = \left\{\theta \in \left(0, \frac{\pi}{2}\right) : \sum_{m=1}^9 \sec\left(\theta + (m-1)\frac{\pi}{6}\right)\sec\left(\theta + \frac{m\pi}{6}\right) = -\frac{8}{\sqrt{3}}\right\}$ Then [JEE (Main) 2022]
 (1) $S = \left\{\frac{\pi}{12}\right\}$ (2) $S = \left\{\frac{2\pi}{3}\right\}$ (3) $\sum_{\theta \in S} \theta = \frac{\pi}{2}$ (4) $\sum_{\theta \in S} \theta = \frac{3\pi}{4}$
 MTE039
15. The number of elements in the set $S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2 + x}{6}\right) = 4^x + 4^{-x}\right\}$ is: [JEE (Main) 2022]
 (1) 1 (2) 3 (3) 0 (4) infinite
 MTE040
16. The number of elements in the set $S = \{\theta \in [0, 2\pi] : 3\cos^4 \theta - 5\cos^2 \theta - 2\sin^6 \theta + 2 = 0\}$ is [JEE (Main) 2023]
 (1) 10 (2) 8 (3) 9 (4) 12
 MTE073
17. If m and n respectively are the numbers of positive and negative values of θ in the interval $[-\pi, \pi]$ that satisfy the equation $\cos 2\theta \cos \frac{\theta}{2} = \cos 3\theta \cos \frac{9\theta}{2}$, then mn is equal to. [JEE (Main) 2023]
 MTE074

18. If the solution of the equation $\log_{\cos x} \cot x + 4 \log_{\sin x} \tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha + \sqrt{\beta}}{2}\right)$, where α, β are integers and $\sqrt{\beta} \notin \mathbb{Q}$ then $\alpha + \beta$ is equal to: [JEE (Main) 2023]
 (1) 3 (2) 5 (3) 6 (4) 4 MTE075
19. Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$, then $\frac{1}{6}(\beta - 14)^2$ is equal to [JEE (Main) 2023]
 (1) 32 (2) 8 (3) 64 (4) 16 MTE076
20. Let $S = \{\theta \in [0, 2\pi) : \tan(\pi \cos \theta) + \tan(\pi \sin \theta) = 0\}$. Then $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to [JEE (Main) 2023]
 MTE077
21. The number of solutions of $\sin^2 x + (2 + 2x - x^2)\sin x - 3(x - 1)^2 = 0$, where $-\pi \leq x \leq \pi$, is [JEE (Main) 2024]
 MTE127
22. Let $|\cos \theta \cos(60 - \theta) \cos(60 + \theta)| \leq \frac{1}{8}, \theta \in [0, 2\pi]$
 Then, the sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value, is : [JEE (Main) 2024]
 (1) 9π (2) 18π (3) 6π (4) 15π MTE128
23. The number of solutions of the equation $4 \sin^2 x - 4 \cos^3 x + 9 - 4 \cos x = 0; x \in [-2\pi, 2\pi]$ is : [JEE (Main) 2024]
 (1) 1 (2) 3 (3) 2 (4) 0 MTE129
24. If $2 \tan^2 \theta - 5 \sec \theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in \mathbb{N}$ then $\sum_{k=1}^n \frac{k}{2^k}$ is equal to : [JEE (Main) 2024]
 (1) $\frac{1}{2^{15}}(2^{14} - 14)$ (2) $\frac{1}{2^{14}}(2^{15} - 15)$ (3) $1 - \frac{15}{2^{13}}$ (4) $\frac{1}{2^{13}}(2^{14} - 15)$ MTE130
25. If $2 \sin^3 x + \sin 2x \cos x + 4 \sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right], n \in \mathbb{N}$, then the roots of the equation $x^2 + nx + (n - 3) = 0$ belong to : [JEE (Main) 2024]
 (1) $(0, \infty)$ (2) $(-\infty, 0)$ (3) $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$ (4) $\mathbb{Z}$ MTE131
26. If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4 \cos \theta + 5 \sin \theta = 1$, then the value of $\tan \alpha$ is [JEE (Main) 2024]
 (1) $\frac{10 - \sqrt{10}}{6}$ (2) $\frac{10 - \sqrt{10}}{12}$ (3) $\frac{\sqrt{10} - 10}{12}$ (4) $\frac{\sqrt{10} - 10}{6}$ MTE132

EXERCISE - JEE (Advanced) PYQ

- For $x \in (0, \pi)$, the equation $\sin x + 2 \sin 2x - \sin 3x = 3$ has
(A) infinitely many solutions (B) three solutions
(C) one solution (D) no solution
[JEE (Advanced) 2014] MTE062
- The number of distinct solutions of equation $\frac{5}{4} \cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is
[JEE (Advanced) 2015] MTE063
- Let $S = \left\{ x \in (-\pi, \pi) : x \neq 0, \pm \frac{\pi}{2} \right\}$. The sum of all distinct solution of the equation $\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to -
[JEE (Advanced) 2016] MTE064
(A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$
- Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a \cos x + 2b \sin x = c$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is ____
[JEE (Advanced) 2018] MTE065

Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order :

$$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$$

$$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}.$$

List-I contains the sets X, Y, Z and W . List -II contains some information regarding these sets.

List-I

- (I) X
(II) Y
(III) Z
(IV) W

List-II

- (P) $\supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$
(Q) an arithmetic progression
(R) NOT an arithmetic progression
(S) $\supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$
(T) $\supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$
(U) $\supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$

- Which of the following is the only CORRECT combination ?
(A) (II), (R), (S) (B) (I), (P), (R) (C) (II), (Q), (T) (D) (I), (Q), (U)
[JEE (Advanced) 2019] MTE066

Trigonometric Equations

6. Which of the following is the only CORRECT combination? [JEE (Advanced) 2019]
 (A) (IV), (Q), (T) (B) (IV), (P), (R), (S) (C) (III), (R), (U) (D) (III), (P), (Q), (U)

MTE067

7. Let $f: [0, 2] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = (3 - \sin(2\pi x)) \sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$$
 [JEE (Advanced) 2020]

If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2]: f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is ____

MTE068

8. Consider the following lists :

List-I	List-II
(I) $\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P) has two elements
(II) $\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3} \tan 3x = 1\right\}$	(Q) has three elements
(III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(R) has four elements
(IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S) has five elements
	(T) has six elements

The correct option is :

[JEE (Advanced) 2022]

- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

MTE069

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	C	C	C	B	A	B	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	B	D	B	A	A	C	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	B,C	A,B,C,D	A,B,C	B,C	B,D	A,C	B,C	A,C,D	C
Que.	11	12	13	14	15					
Ans.	C	A	B	A	A → S; B → P; C → R; D → Q					

EXERCISE (S)

1.	3	2.	3	3.	24	4.	07	5.	17
6.	2	7.	5025	8.	50	9.	6	10.	5

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	4	1	1	1	2	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	2	3	1	3	25	4	1	2
Que.	21	22	23	24	25	26				
Ans.	2	3	4	4	2	3				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8		
Ans.	D	8	C	0.50	C	B	1.00	B		